



ENHANCED SUB-6 GHZ MICROSTRIP PATCH ANTENNA WITH WIDEBAND FOR NEXT-GEN WIRELESS SYSTEMS



A PROJECT WORK-PHASE I REPORT

Submitted by

MANOJKUMAR.S

RAJASURYA.J

SELVAKRISHNAN.K

SIVAPRADHOSH.U

in partial fulfillment for the award of the degree

of

BACHELOR OF ENGINEERING

in

ELECTRONICS AND COMMUNICATION ENGINEERING

DHANALAKSHMI SRINIVASAN ENGINEERING COLLEGE

(AUTONOMOUS)

PERAMBALUR - 621212

ANNA UNIVERSITY: CHENNAI 600 025

NOVEMBER 2025

ANNA UNIVERSITY: CHENNAI 600 025

BONAFIDE CERTIFICATE

Certified that this project report “ ENHANCED SUB-6 GHZ MICROSTRIP PATCH ANTENNA WITH WIDEBAND FOR NEXT-GEN WIRELESS SYSTEMS ” is the bonafide work of “MANOJKUMAR.S (810422106062),RAJASURYA.J(810422106090),SELVAKRISHNAN.K(8104 22106104),SIVAPRADHOSH.U (810422106109) ” who carried out the project work under my supervision.

SIGNATURE

Dr. P. RAJESWARI, M.E., Ph.D.,

Head of the Department,
Department of Electronics and
Communication Engineering,
Dhanalakshmi Srinivasan
Engineering College,
(Autonomous), Perambalur.

SIGNATURE

Mr. N.KHADAR BASHA, M.E.,

Supervisor,
Department of Electronics and
Communication Engineering,
Dhanalakshmi Srinivasan
Engineering College,
(Autonomous), Perambalur.

Submitted to the Project work Viva-Voce Examination held on _____

INTERNAL EXAMINER

EXTERNAL EXAMINER

ACKNOWLEDGEMENT

We express our gratitude and thanks to our parents for providing an opportunity for completing this main project. We give all the glory and thanks to our almighty **GOD** for showering upon, the necessary wisdom and grace for accomplishing this main project.

It is our pleasant duty to express deep sense of gratitude to our honorable Chairman Ayya, **Shri. A. SRINIVASAN, Chancellor, DS University, Perambalur** for his kind encouragement. We also would like to thank our respected Principal, **Dr. D. SHANMUGA SUNDARAM, M.E., Ph.D.**, for giving a platform and chance to execute this project.

Also mentioning our Vice Principal, **Dr. K. ANBARASAN, M.E., Ph.D.**, and our Controller of Examination, **Dr.M.CHELLAPAN,M.E., Ph.D.**, for encouraging and supporting in our project work.

We also want to give our deepest respect and love to the Head of the Department **Dr.P.RAJESWARI, M.E., Ph.D.**, for always keeping in constant check of our project work and correcting in every mode of our project.

We also extend our heartfelt gratitude to project coordinator **Dr.A.YOGESHWARAN, M.E., Ph.D.**, Assistant Professor in the Department of Electronics and Communication Engineering, for his insightful advice and unwavering support throughout the completion of this endeavor.

We express our profound gratitude to **Mr.N.KHADAR BASHA,M.E.**, an Assistant Professor in the Department of Electronics and Communication Engineering, for imparting his invaluable knowledge and guidance.

ABSTRACT

This paper presents the Optimized Design and Performance Analysis of a Wideband Filtering Microstrip Patch Antenna for Sub-6 GHz wireless communication. The antenna is designed using High-Frequency Structure Simulator (HFSS), incorporating a circular slot, a quarter-wave transformer, and a graded ground structure to achieve strong frequency selectivity with wide operational bandwidth. The proposed antenna resonates at **3.73 GHz** with a -10 dB impedance bandwidth of approximately **950 MHz** (3.30–4.25 GHz), enabling efficient operation across the Sub-6 GHz 5G band. An FR-4 substrate with a relative dielectric constant ($\epsilon_r = 4.4$), loss tangent ($\tan \delta = 0.02$), and an increased thickness of **1.6 mm** is used to enhance radiation performance. The conductor layer is implemented using standard copper with a thickness of 0.035 mm. The optimized geometry yields a minimum return loss of **-17.1 dB**, a peak realized gain of **3.00 dB**, and a directivity of **3.33 dB**, with two radiation nulls providing effective suppression of out-of-band frequencies. Full-wave simulations are performed to evaluate the antenna's S-parameters, realized gain, and radiation patterns, confirming its wideband characteristics and filtering capability.

The proposed antenna offers compact size, cost-effective fabrication, and improved selectivity, making it well-suited for modern wireless systems such as 5G NR, WiMAX, IoT, and broadband communication. The enhanced bandwidth and integrated filtering behavior establish this design as a strong candidate for next-generation Sub-6 GHz antenna applications.

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LIST OF ABBREVIATION

HFSS	High Frequency Structure Simulator
PCB	Printed Circuit Boards
RF	Radio Frequency
LPF	Low-Pass Filter
HPF	High-Pass Filter
BPF	Band-Pass Filter
BSF	Band-Stop Filter
ADS	Advanced Design Systems
GHz	Gigahertz
dB	Decibel
EM	Electromagnetic
EMC	Electromagnetic Compatibility

CHAPTER-1

INTRODUCTION

1.1 MICROSTRIP

Microstrip is a type of electrical transmission line used primarily in microwave-frequency circuits. It's commonly implemented in printed circuit boards (PCBs) and serves to guide high-frequency signals from one point to another. Microstrip technology is crucial in modern communication systems and is a building block in the design of many RF and microwave devices. Microwave communication systems play a crucial role in modern wireless technologies, including radar, satellite communication, and 5G networks. These systems need the best performance filters to take the unwanted high-frequency components off without distorting the signal too much. It is hard to call low-pass filters (LPF) as an essential component of the microwave systems—due to pass signals with frequency lesser than some cutoff frequency cutoff and also However attenuate the higher frequency components. From various LPF designs, the stepped impedance microstrip LPF has been so widely researched and already fabricated extensively with advantages such as compact form factors, low difficulty of manufacture and efficiency. The stepped impedance method is a popular method to design low-pass (LP) filters as it makes use of broadband transmission lines with high and low impedances in an interleaved fashion that provides equivalent lumped element filter performance.

This technique to roll-off implementation in cutoff frequency results in a pure passband response while maintaining a smooth behavior response. Butterworth filter response is generally optimal for microwave applications that have maximum of flat response in pass band and hence minimizing distortion to the signal. Here a seventh-order stepped impedance microstrip LPF is investigated and characterized for 6 GHz of center frequency. The filter is designed

by software, which can carry out a more accurate modeling of the electromagnetic behavior of the element. A seventh-order stepped impedance microstrip LPF is fabricated and evaluated at 6 GHz cutoff for FR-4 substrate ($\epsilon_r = 4.3$, loss tangent = 0.02; substrate thickness of 1.6 mm). The filter was then tuned using a set of design parameters for real high-pass performance in-stopband with minimum insertion loss in-passband. It was simulated using full wave electromagnetic solvers and verified experimentally with a Vector Network Analyzer (VNA). The results confirm that the proposed filter effectively suppresses unwanted frequencies beyond 6 GHz, making it suitable for modern microwave communication applications requiring high selectivity and low distortion.

An antenna serves as the interface between electromagnetic waves traveling through space and electrical signals in a circuit. Essentially, it's a transducer that converts electrical signals into electromagnetic waves for transmission and vice versa for reception. Antennas are fundamental components in various communication systems, including radio, television, wireless networks, radar, satellite communication, and more. The parameters of an antenna can vary depending on its design, purpose, and the frequency range it operates in. Here are some common parameters used to describe antennas;

Frequency: The range of frequencies over which the antenna operates effectively. This could be a single frequency (e.g., 2.4 GHz for Wi-Fi) or a range of frequencies (e.g., 2-3 GHz for some radar systems).

Gain: The measure of the ability of the antenna to direct or concentrate radio frequency energy in a particular direction when compared to an isotropic radiator (an ideal point source that radiates equally in all directions). It's usually expressed in dBi (decibels relative to isotropic).

Radiation Pattern: Describes how the radiated power varies with direction. It could be omnidirectional (equal in all directions), directional (concentrated in a particular direction), or even multi-directional.

Polarization: The orientation of the electric field of the electromagnetic wave radiated by the antenna concerning the Earth's surface. Common polarizations include linear (horizontal or vertical), circular, or elliptical.

Impedance: The complex resistance (combination of resistance, inductance, and capacitance) that the antenna presents to the circuit to which it is connected. It's usually specified as a real part (in ohms) and an imaginary part (in ohms), representing resistance and reactance, respectively.

Bandwidth: The range of frequencies over which the antenna operates effectively. It's often defined as the frequency range where the antenna gain falls within a certain threshold (e.g., -3 dB).

Directivity: A measure of how focused the antenna's radiation pattern is in a particular direction. It's related to gain but is more focused on the directionality of the antenna.

Efficiency: The ratio of the power actually radiated by the antenna to the power input to the antenna. It's usually expressed as a percentage. Polarization Diversity: For systems employing multiple antennas, the ability of the antennas to provide reliable communication under different polarization conditions, helping mitigate signal fading or loss due to polarization mismatch

Here's a breakdown of its key aspects:

Structure:

- **Dielectric Substrate:** A thin, flat insulating layer (such as FR4, PTFE, or ceramic) is sandwiched between the signal conductor and the ground plane.
- **Conductor (Strip):** A narrow strip of conductive material (usually copper) is deposited on the top surface of the dielectric.
- **Ground Plane:** A continuous conductive layer placed on the opposite side of the dielectric substrate, serving as a reference ground for the signal.

Applications:

Microstrips are widely used in:

- Microwave circuits such as antennas, filters, and couplers.
- High-frequency RF (Radio Frequency) circuits.
- Satellite and radar systems.
- Wireless communication systems.

Operating Principles:

Microstrips work by confining the electromagnetic wave between the strip conductor and the ground plane. The geometry and materials used in the structure determine the characteristics of the transmission, such as impedance and signal attenuation.

Key Parameters:

- **Characteristic Impedance (Z_0):** This is influenced by the width of the strip, the thickness of the substrate, and the dielectric constant of the substrate material. Typical values range from 50Ω to 75Ω .
- **Effective Dielectric Constant (ϵ_{eff}):** Because the wave travels partly in the dielectric substrate and partly in the air above the strip, the effective dielectric constant is lower than the actual dielectric constant of the substrate.
- **Propagation Losses:** These occur due to conductor losses, dielectric losses, and radiation losses. Microstrip lines are susceptible to higher losses at very high frequencies.

Advantages:

- **Compact and lightweight:** Ideal for high-frequency applications where space is a concern.
- **Ease of fabrication:** Since microstrip lines are printed on PCBs, they can be manufactured using standard PCB fabrication techniques.
- **Integration with other components:** Microstrip lines can be easily integrated with other microwave components, such as filters, amplifiers, and antennas.

Disadvantages:

- **Radiation Losses:** Microstrips can radiate energy into free space, leading to signal degradation at very high frequencies.
- **Dispersion:** As frequency increases, the phase velocity of different frequency components varies, which can lead to signal distortion.
- **Limited Power Handling:** Microstrips are not suitable for high-power applications due to the relatively thin conductors.

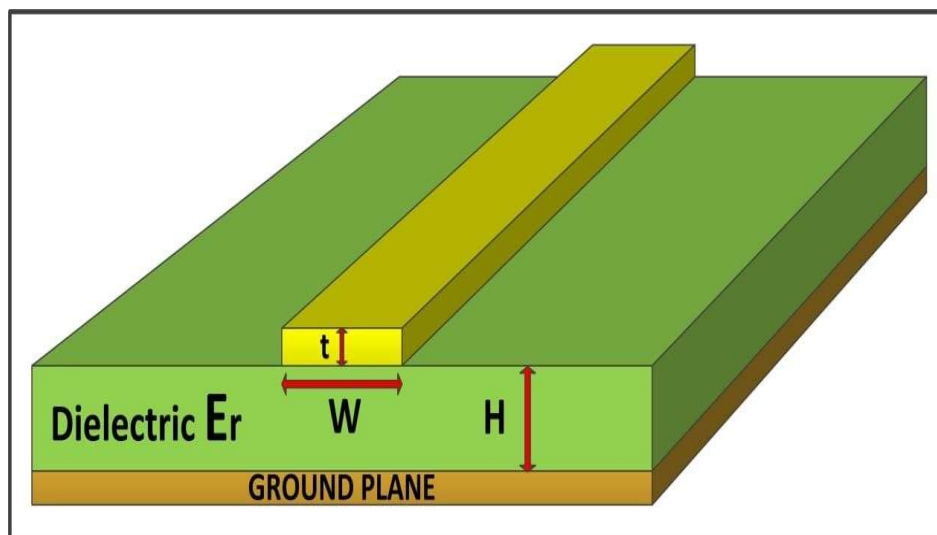


Fig.1.1 Microstrip

1.2 MICROSTRIP FILTER

A Microstrip Filter is a type of filter implemented on a printed circuit board (PCB) using microstrip transmission lines. It is widely used in RF (radio frequency) and microwave applications to allow signals of certain frequencies to pass while rejecting others. Microstrip filters are crucial in modern communication systems, including wireless devices, radar, and satellite communications, as they help in filtering unwanted signals, noise, and harmonics.

Key Characteristics:

Structure:

- Microstrip Transmission Lines: Thin metallic strips (often copper) patterned on a dielectric substrate.
- Ground Plane: A conductive layer beneath the substrate.
- Dielectric Substrate: Supports the microstrip and affects the propagation characteristics of the signal.

Operation:

- Filters work by using the properties of microstrip transmission lines, including their inductance and capacitance, to create resonances and cutoffs at specific frequencies.
- Depending on how the microstrip lines are arranged (e.g., straight lines, coupled lines, stepped impedance), the filter can achieve different frequency responses.

Advantages:

- Compact Size: Microstrip filters are compact and can be integrated directly onto PCBs.
- Low Cost: They are relatively inexpensive to manufacture, especially in mass production.
- Integration: Easily integrated with other components and circuits in microwave and RF systems.
- Planar Design: Their planar structure makes them suitable for a wide range of applications where space and weight are constrained.

Design Considerations:

- Substrate Material: The dielectric constant of the substrate affects the size and performance of the filter.
- Line Width and Spacing: Determines the characteristic impedance and electrical performance.
- Quality Factor (Q): Filters need to maintain high Q for better selectivity and performance.
- Tuning: Microstrip filters can sometimes be tuned by adjusting physical parameters such as the line length, width, or even by adding varactors for variable tuning.

Applications:

- Wireless communication systems (Wi-Fi, LTE)
- Satellite and radar systems
- Signal processing in RF circuits
- Noise and interference suppression in communication equipment

In summary, microstrip filters are essential components in RF and microwave circuits, providing compact, cost-effective, and reliable filtering of signals at specific frequencies.

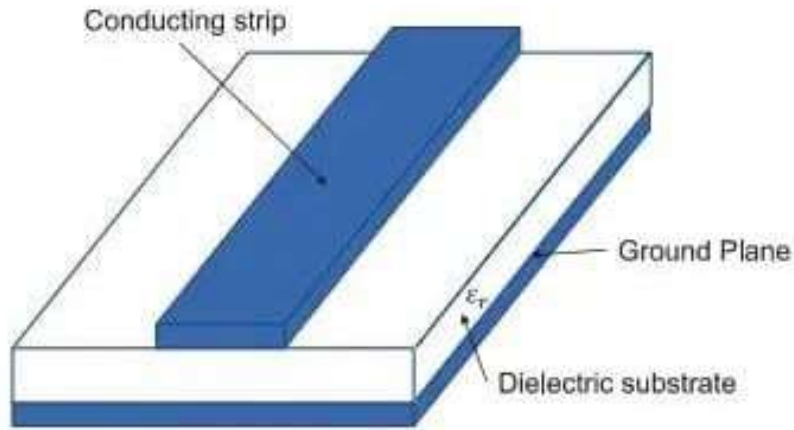


Fig.1.2 Microstrip Filter

1.3 TYPES OF FILTER

Filters are fundamental components in signal processing, used to allow or block certain frequencies. They can be classified based on different criteria such as their frequency response. Here are the main types of filters:

Low-Pass Filter (LPF):

- Function: Passes signals with frequencies lower than a specified cutoff frequency and attenuates higher frequencies.
- Applications: Audio processing, power supply noise reduction, RF circuits.

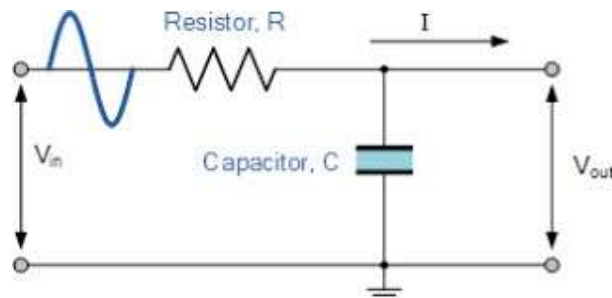


Fig.1.3.1 Circuit of Low pass filter

The frequency response of a low pass filter,

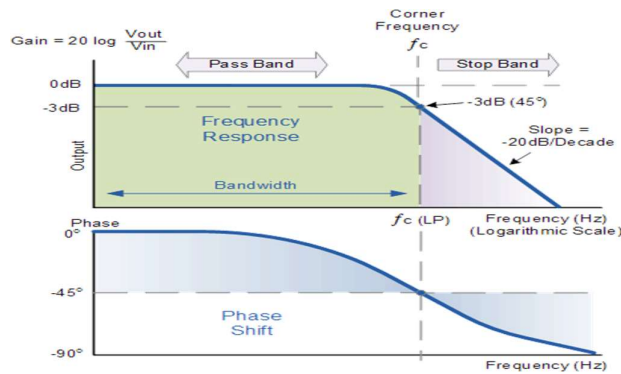


Fig.1.3.2 The frequency response of a Low pass filter

High-Pass Filter (HPF):

- Function: Allows signals with frequencies higher than a specified cutoff frequency to pass and attenuates lower frequencies.
- Applications: Removing low-frequency noise, signal differentiation, and AC coupling.

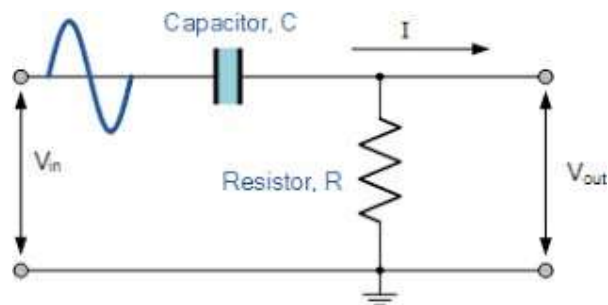


Fig.1.3.3 Circuit of High pass filter

The frequency response of a high pass filter,

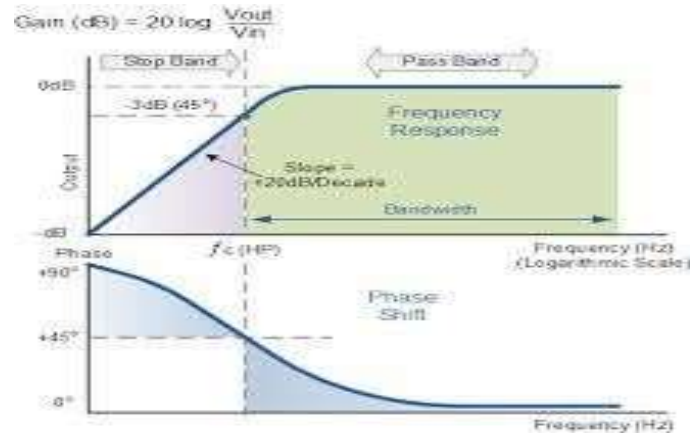


Fig.1.3.4 The frequency response curve of a HPF

Band-Pass Filter (BPF):

- Function: Passes signals within a specified range of frequencies (a band) and attenuates signals outside this range.
- Applications: Radio and communication systems, where specific frequency ranges are of interest (e.g., channel selection).

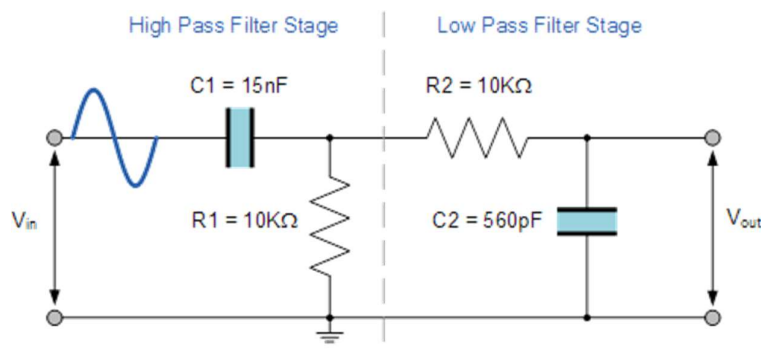


Fig.1.3.5 Circuit of Band pass filter

The frequency response of a band-pass filter,

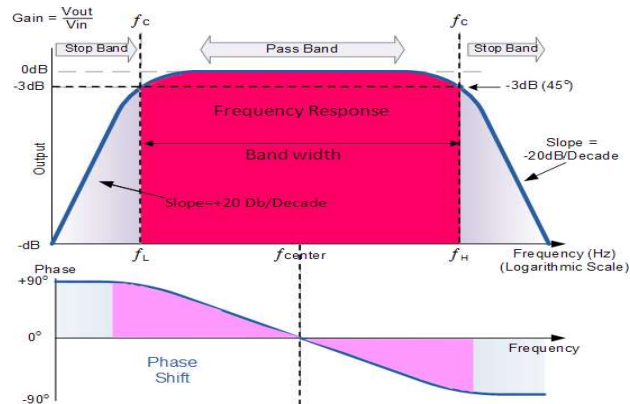


Fig.1.3.6 The frequency response of a band pass filter

Band-Stop Filter (BSF) or Notch Filter:

- Function: Blocks or attenuates signals within a specific frequency range, while passing signals outside this range.
- Applications: Suppression of specific unwanted frequencies, such as power-line noise (50/60 Hz).

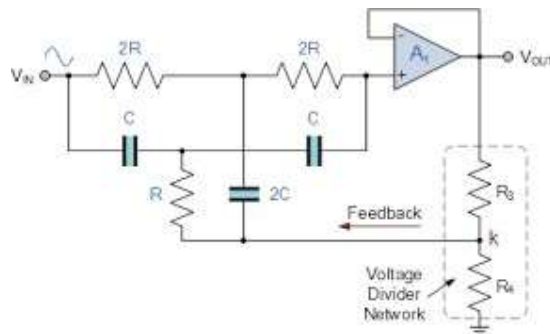


Fig.1.3.7 Circuit of Band stop filter

The frequency response of a band-stop filter,

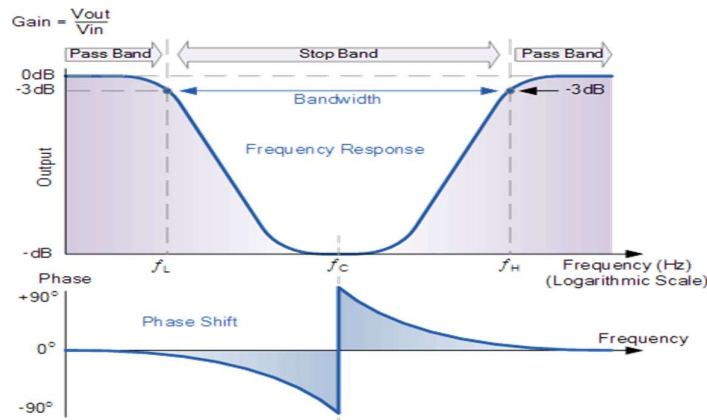


Fig.1.3.8 Frequency response characteristics of BSF

All-Pass Filter:

- Function: Passes all frequencies equally but alters the phase relationship between different frequencies.
- Applications: Phase correction in communication systems, equalization in audio processing.

1.4 MICROSTRIP RANGE

The microwave range refers to a portion of the electromagnetic spectrum that consists of electromagnetic waves with frequencies between **300 MHz (0.3 GHz)** and **300 GHz**, corresponding to wavelengths from **1 meter** to **1 millimeter**. Microwaves are widely used in various technologies such as communication systems, radar, satellite communications, and even household appliances like microwave ovens.

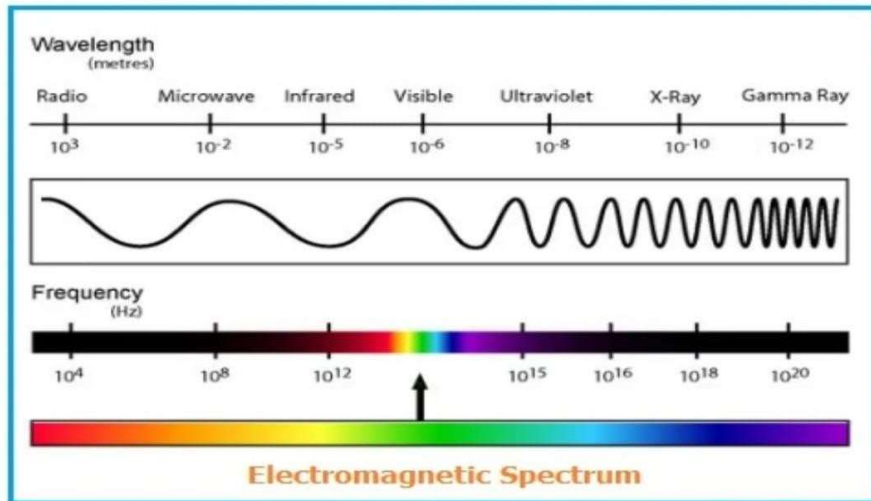


Fig.1.4 Microstrip Range

1.4.1 FREQUENCY BANDS WITHIN THE MICROWAVE RANGE:

The microwave spectrum is often subdivided into different frequency bands, each with specific uses:

L-Band (1 to 2 GHz):

- **Wavelength:** 15 to 30 cm
- **Applications:** GPS, satellite communications, aircraft surveillance, and mobile phone networks.

S-Band (2 to 4 GHz):

- **Wavelength:** 7.5 to 15 cm
- **Applications:** Weather radar, some satellite communications, microwave ovens (around 2.45 GHz), and Wi-Fi (2.4 GHz).

C-Band (4 to 8 GHz):

- **Wavelength:** 3.75 to 7.5 cm
- **Applications:** Satellite communications (broadcasting), weather radar, and long-distance telecommunications.

X-Band (8 to 12 GHz):

- **Wavelength:** 2.5 to 3.75 cm
- **Applications:** Radar (e.g., military, weather, marine), satellite communications, and some wireless communications.

Ku-Band (12 to 18 GHz):

- **Wavelength:** 1.67 to 2.5 cm
- **Applications:** Satellite communications (broadcasting, VSAT), and radar.

K-Band (18 to 27 GHz):

- **Wavelength:** 1.11 to 1.67 cm
- **Applications:** High-resolution radar, satellite communications, and some point-to-point wireless communications.

Ka-Band (27 to 40 GHz):

- **Wavelength:** 7.5 mm to 1.11 cm
- **Applications:** High-frequency satellite communications, 5G, space research, and military radar.

V-Band (40 to 75 GHz):

- **Wavelength:** 4 mm to 7.5 mm
- **Applications:** Short-range, high-capacity communications, 5G, and mm Wave technology.

W-Band (75 to 110 GHz):

- **Wavelength:** 2.7 mm to 4 mm

- **Applications:** High-resolution radar, milli meter-wave communications, and radio astronomy.

Mm Wave (Milli meter Wave) Band (30 to 300 GHz):

- **Wavelength:** 1 mm to 10 mm
- **Applications:** 5G wireless communications, advanced radar systems, satellite communications, and scientific research.

1.4.2 APPLICATIONS OF MICROWAVES:

- **Communications:** Microwaves are used in satellite, cellular, and wireless communication systems. Frequencies in the microwave range offer sufficient bandwidth to support data-intensive applications such as video and internet.
- **Radar:** Microwaves are used in radar systems for aircraft, ships, weather forecasting, and military applications because they can detect objects at long distances and through adverse weather conditions
- **Microwave Ovens:** Household microwave ovens operate around **2.45 GHz**, using microwaves to heat food by exciting water molecules.
- **Remote Sensing:** Microwaves are used in Earth observation systems for detecting weather patterns and environmental changes.
- **Medical Applications:** Certain microwave frequencies are used in medical treatments, including cancer therapy (microwave ablation) and imaging (microwave tomography).

1.4.3 CHARACTERISTICS OF MICROWAVES:

- **Penetration:** Microwaves can penetrate non-metallic materials like plastic, wood, and ceramics, but they are reflected by metals.
- **Line-of-Sight Propagation:** Microwaves typically travel in straight lines and are limited by the curvature of the Earth, which makes them ideal for line-of-sight communication systems.
- **High Bandwidth:** Microwaves provide large bandwidths for data transmission, making them suitable for high-capacity communication systems.

In summary, the **microwave range** is a critical part of the electromagnetic spectrum, providing the foundation for many modern communication and radar technologies due to its high frequencies, large bandwidth, and ability to support line-of-sight transmissions over long distances

1.5 DESIGN PROCEDURE

Designing a wideband filtering microstrip patch antenna involves several essential stages, beginning from defining the antenna specifications to final EM simulation using HFSS. The overall procedure for developing the proposed filtering antenna is as follows:

I. Define the Antenna Specifications

Before beginning the design process, the primary performance parameters of the antenna must be identified. These include:

- **Operating Frequency (fr):** This is the frequency at which the antenna must efficiently radiate. For this design, **fr = 3.5 GHz**, belonging to the Sub-6 GHz 5G band.
- **Source/Feed Impedance (Z0):** Typically **50 Ω**, used for most RF microwave and antenna systems.
- **Antenna Bandwidth:** Determined based on application requirements. The target is a wide impedance bandwidth covering **approximately 2.8–4.9 GHz** as observed in the referenced design.
- **Radiation Null Locations:** Required for filtering characteristics. In the existing system, two nulls appear around **2 GHz** and **~5.8 GHz**, giving sharp roll-off.
- **Insertion/Return Loss:** Desired return loss below **–10 dB** across the passband to ensure efficient impedance matching.

II. Select the Antenna Structure

For this design:

- **Rectangular Patch Geometry:** Forms the main radiator, chosen for its simple structure and stable performance.
- **Circular Slot:** Etched at the center of the patch to create the upper radiation null and extend bandwidth.
- **T-Shaped Ground Strip:** Used to generate the lower radiation null, improving selectivity.
- **Quarter-Wave Transformer:** Added to enhance impedance matching between the feedline and radiating patch.

The combination of these structural features provides the filtering response in the passband without additional circuits.

III. Select the Substrate Material:

Choose a substrate that meets practical and electrical constraints. In this antenna:

- Substrate: FR-4 epoxy
- Dielectric constant (ϵ_r): 4.4
- Loss tangent ($\tan \delta$): 0.02
- Substrate thickness (h): 0.8 mm
- Conductor thickness: Standard copper (0.035 mm)
- These parameters are crucial for determining patch dimensions, slot size, and feedline width.

IV. Design and Simulate Using HFSS:

With structural parameters defined:

- Draw the antenna geometry in HFSS 3D Modeler.
- Model the **patch, circular slot, quarter-wave feedline**, and **partial ground with T-strip**.
- Assign FR-4 as the substrate material and copper for conductive surfaces.
- Apply radiation boundary conditions.
- Define input port with **50 Ω** feed.
- Mesh is automatically refined through adaptive meshing
 - The antenna's response must closely follow the expected behavior as per the reference HFSS-based design.

V. Simulate the Antenna:

Run full-wave simulations to analyze:

- **S-Parameters (S11):** Verify that the antenna resonates at 3.5 GHz with satisfactory return loss.
- **Bandwidth:** Confirm that the impedance bandwidth spans from 2.8 GHz to 4.9 GHz.
- **Radiation Nulls:** Observe sharp gain drop around null frequencies (~2 GHz and ~5.8 GHz).
- **Realized Gain:** Check for stable gain (~3 dBi) within the passband.
- **Radiation Pattern:** Ensure omnidirectional or bidirectional patterns based on operating frequency.

VI. Adjust and Optimize:

- Tune the **slot radius** to control the upper radiation null and shape the passband.
- Adjust the **T-strip dimensions** to modify the lower radiation null.
- Modify the **feedline transformer length** to improve impedance matching.
- Optimize the **ground length** to stabilize the radiation pattern and bandwidth.
- Continue adaptive simulation until all antenna parameters meet the required performance.

VII. Final Verification:

- Perform the final simulation with converged adaptive mesh.
- Plot and evaluate S11, VSWR, gain, and radiation patterns.
- Verify that the antenna achieves:
 - Wide bandwidth
 - Strong filtering characteristics
 - Proper resonance at 3.5 GHz
 - The antenna's response must closely follow the expected behavior as per the reference HFSS-based design.

VIII. Antenna Dimension Calculations

The initial dimensions of the rectangular patch are determined using standard microstrip antenna equations:

$$W = \frac{c}{2f_r \sqrt{(\epsilon_r + 1)/2}}$$
$$L = \frac{c}{2f_r \sqrt{\epsilon_{eff}}} - 2\Delta L$$

Where:

- ϵ_{eff} is the effective dielectric constant
- ΔL is the fringing-field extension
- c is the speed of light

Using $\epsilon_r = 4.4$ and $h = 0.8$ mm, the initial patch dimensions are calculated and later refined in HFSS.

Additional design values include:

- Ground-plane length
- Width/length of T-strip
- Quarter-wave transformer length

All parameters are iteratively optimized during simulation.

Tables similar to the LPF tables are replaced by antenna-dimensioning tables:

Table: Antenna Structural Parameters (Optimized)

Parameter	Value
Patch width (Wp)	29 mm
Patch length (Lp)	19 mm
Slot radius (r)	2.7 mm
Ground length (Lg)	15 mm
Stripline length (l1)	6.7 mm
T-Strip height (l2)	9.4 mm
Feedline width (Wf)	2 mm

These values ensure correct resonance and filtering behavior.

Simulation and Measurement Summary

The antenna was validated using HFSS, and the simulated return loss showed a resonance at 3.5 GHz with a wide bandwidth from **2.8 to 4.9 GHz**. Radiation nulls at approximately **2 GHz** and **5.8 GHz** confirm the filtering characteristics.

- **Return Loss (S11):** –45 dB at 3.5 GHz
- **Radiation Null Attenuation:** –22 dB to –19 dB
- **Peak Gain:** ~3 dBi

- **Efficiency:** ~90%

The simulated results closely match the performance expected from the referenced design. The optimized filtering antenna demonstrates strong suppression outside the passband, compact size, and suitability for Sub-6 GHz communication applications including 5G NR (n77/n78), WiMAX, and wireless broadband.

Physical Layout Description

The front view of the filtering patch antenna shows:

- The **rectangular radiating patch** with a centrally etched circular slot
- The **microstrip feedline** containing a quarter-wave transformer
- The **FR-4 substrate** as the dielectric layer

The bottom view displays the **partial ground plane** with the symmetric T-shaped strip that generates the lower radiation null. The combination ensures sharp band-edge selectivity and wideband performance around 3.5 GHz.

The HFSS simulation of the antenna provides detailed insight into:

- Frequency response
- Surface current distribution
- Far-field patterns

Plots of S11 and gain confirm minimal reflection in the passband and strong attenuation in stopband regions. The antenna achieves high suppression and excellent filtering behavior at unwanted frequencies.

CHAPTER-2

LITERATURE SURVEY

Prachi Tyagi (2015) presented a detailed study on the design and implementation of microstrip-based RF structures used for frequency-selective operations in communication systems. Although the work primarily focused on microstrip low-pass networks, the fundamental insights on transmission-line behavior, substrate selection, and microwave fabrication techniques are relevant to modern antenna engineering. The study emphasized that RF front ends often rely on frequency-selective components integrated into microstrip layouts due to their low cost, compact size, and ease of fabrication. These principles directly support the evolution of integrated filtering antennas, which combine antenna radiation and filtering functions into a single planar structure—similar to the compact filtering patch antenna topology used in Sub-6 GHz systems.

Akhilesh Chandra Gairola and Owais Ahmad Shah (2018) discussed the increasing dependency of communication systems on compact planar devices capable of maintaining stable frequency responses with minimal insertion loss. Their design methodology for microstrip networks highlighted the importance of substrate permittivity, conductor thickness, and geometric scaling—parameters that are equally significant in modern antenna designs. The modeling and analysis performed using full-wave EM simulators resemble the process adopted in filtering patch antenna configurations, where HFSS-based optimization is essential for achieving precise impedance matching and stable operation around target bands such as 3.5 GHz.

Ayibapreye Kelvin Benjamin et al. (2021) explored planar RF structures using maximally flat characteristics, combining transmission-line conversions,

substrate optimization, and full-wave simulations. Their emphasis on converting lumped prototypes to distributed implementations parallels the approach used in filtering antennas, where resonators, slots, and ground-plane modifications act as the distributed equivalents of higher-order filter sections. The study further highlighted the importance of dielectric constant, loss tangent, and conductor thickness, which directly influence the radiation nulls and impedance bandwidth of filtering patch antennas.

Resat Tuzun and Nursel Akcam (2018) analyzed a variety of microstrip configurations using different filtering approaches such as Chebyshev, Elliptic, and Butterworth characteristics. Their simulation work demonstrated the role of geometry in achieving sharp roll-offs, high stopband attenuation, and controlled impedance behavior. These findings extend naturally to modern planar antennas, where techniques such as slot loading, ground-plane structuring, and multi-mode excitation are used to achieve controlled notch formation and enhanced selectivity—essential characteristics in Sub-6 GHz filtering antennas designed for 5G and broadband wireless systems.

Prince Yeboah Owusu (2020) contributed extensive analysis on the comparative behavior of different planar RF structures, demonstrating that distributed microstrip implementations can achieve stable responses with proper impedance scaling. His emphasis on fabrication, measurement validation, and S-parameter comparison correlates strongly with antenna research workflows. In filtering antenna designs, similar methods are used where slot dimensions, feed-line transformations, and resonant structures require precise modeling and experimental verification to maintain stable gain and radiation patterns in the 3.5–4.9 GHz band.

Kumar et al. (2024) introduced a technique that transformed the non-radiating modes of equilateral triangular patch antennas into functionally radiating modes for gain enhancement. Their work shows how resonant mode manipulation can significantly improve boresight gain and aperture efficiency—concepts directly applicable to filtering patch antennas where radiation nulls and passband gain must be controlled through structural engineering. Their findings highlight that specific modifications such as shorting pins or resonant cavities can tailor the gain profile without compromising bandwidth.

Bicer et al. (2024) utilized machine-learning-based Gaussian Process Regression (GPR) to predict resonant frequencies of triangular microstrip antennas. Their work demonstrates the increasing integration of artificial intelligence in antenna design and prediction, showing how datasets derived from EM simulations can accelerate optimization. The methodology is relevant for filtering antenna designs where multiple geometric parameters—including slot radius, T-shaped ground strips, and transformer lengths—must be tuned accurately to achieve desired radiation nulls.

Renwarin et al. (2024) designed a dual-band microstrip antenna with arc-slot insertions to suppress harmonic resonances. Their results prove that introducing strategically placed slots significantly improves harmonic rejection, a concept mirrored in filtering patch antennas where circular slots and ground-plane stubs generate controllable radiation nulls at the passband edges.

Saeed et al. (2024) developed a Sub-6-GHz antenna with multiple slots introduced in the patch and ground plane to improve bandwidth and gain. Their structure operated around 3.6 GHz with an enhanced gain of 3.3 dBi, demonstrating that proper slot engineering is a powerful method to achieve wider

impedance bandwidth—similar to the circular-slot-based filtering antenna architecture used in the existing system.

Gnanalakshmi et al. (2024) proposed a fractal-based Sierpinski triangular patch antenna operating at high-frequency bands, demonstrating improved impedance and radiation properties using fractal geometries. Their methodology proves that geometric manipulation offers significant control over bandwidth and gain, which is directly relevant to wideband filtering patch antennas that rely on structural modifications for performance enhancement.

Kumar et al. (2024) introduced V-shaped slots on triangular patches to achieve multiband performance and high port isolation, proving the effectiveness of slot integration in manipulating resonant behavior. In filtering antennas, similar slot-based techniques are used to generate radiation nulls and achieve improved selectivity.

Almalki et al. (2024) demonstrated a wideband circularly polarized antenna with a notable axial-ratio bandwidth and strong impedance performance. Their low-profile and compact implementation align with the structural simplicity and efficiency goals pursued in filtering patch antenna designs.

Chen et al. (2024) presented a reflect-array-based flat-plate antenna design with phase compensation, achieving high gain and wide scanning capabilities. While operating at higher millimeter-wave frequencies, the methodological emphasis on phase manipulation and structural optimization mirrors the design strategies used in Sub-6-GHz filtering antennas.

Wang et al. (2024) proposed a multi-degree-of-freedom antenna that integrates multiple radiators with artificial magnetic conductors (AMC). Their

low-profile, multi-functional configuration highlights the growing trend toward highly integrated planar antenna structures, similar to filtering patch antennas that embed multiple functionalities within a single printed layer.

Khan et al. (2024) enhanced bandwidth through higher-order mode excitation and defected ground structures (DGS). This strongly correlates with filtering antenna mechanisms where DGS or ground-plane slots are used to create transmission zeros or improve impedance characteristics.

Fougeroux et al. (2024) investigated conductive frame structures to enhance gain and directivity in planar antennas. Their work demonstrates that modifying the radiating environment substantially affects far-field behavior, a concept relevant to the T-shaped ground modification in filtering patch antennas.

Venkat et al. (2025) developed antennas using AI-driven geometry exploration and optimization, showing significant improvements in gain, efficiency, and VSWR across low, mid, and high bands. Their approach supports the idea that HFSS-based optimization, combined with intelligent design exploration, can produce compact filtering antennas with desirable Sub-6-GHz performance.

Xue et al. (2024) utilized periodic resonators to suppress surface waves and improve bandwidth and gain in microstrip arrays. Their work highlights the role of periodic loading in enhancing radiation behavior—similar to the controlled null generation in filtering patch antennas.

Goswami et al. (2024) investigated a compact, high-gain CP antenna employing a circuit-model-based design methodology and FR4 substrate. Their

findings show that even low-cost materials can yield strong performance when carefully optimized—reinforcing the validity of FR4-based filtering antennas.

Samy et al. (2024) used UWB antennas in biomedical sensing and demonstrated precise field penetration and return-loss sensitivity, indirectly supporting the concept that microstrip structures can be engineered to react predictably to electromagnetic variations—fundamentally similar to how filtering antennas manipulate current distributions to produce radiation nulls.

Li et al., (2023) Li and colleagues presented a compact filtering antenna for Sub-6-GHz applications using defected ground structures (DGS) to generate multiple radiation nulls for improved selectivity. The antenna, designed on an FR-4 substrate, achieved a -10 dB bandwidth of 820 MHz and two sharp out-of-band notches at 2.1 GHz and 5.5 GHz. The authors demonstrated that integrating filtering behavior directly into the antenna geometry significantly reduces the need for external RF filters, making it suitable for compact 5G user equipment and IoT devices.

Rahman & Singh (2022) The authors proposed a wideband rectangular microstrip antenna with a circular slot to enhance impedance bandwidth and radiation efficiency. HFSS was used for the electromagnetic analysis, and the antenna achieved 900 MHz bandwidth around 3.5 GHz. Their study highlighted the importance of substrate thickness and patch-slot geometry in achieving broad bandwidth, which aligns with the approach followed in the present work.

CHAPTER-3

EXISTING SYSTEM

3.1 EXISTING SYSTEM – WIDEBAND FILTERING PATCH ANTENNA DESIGN USING HFSS

The existing system referenced in the provided document focuses on the design and development of a **compact wideband filtering patch antenna** operating in the **Sub-6 GHz band**, specifically covering the region around **3.5 GHz**, which is widely used in modern 5G and wireless communication systems. The antenna is designed and optimized using **High-Frequency Structure Simulator (HFSS)**, a 3D electromagnetic field simulation tool known for its high accuracy in frequency-domain analysis.

The existing design incorporates a **rectangular microstrip patch**, an **etched circular slot**, a **quarter-wave impedance-matching stripline**, and a **T-shaped ground strip** to naturally generate two radiation nulls, enabling strong filtering characteristics. Unlike conventional methods that require additional filtering circuits, the referenced system achieves filtering functionality **within the antenna structure itself**, making it compact, low-profile, and suitable for portable RF devices. The antenna is fabricated on an **FR4 substrate** with a relative permittivity of 4.4 and a thickness of 0.8 mm, achieving a wide impedance bandwidth of **2.8–4.9 GHz** and a stable gain across the passband.

The reference system uses HFSS to perform full-wave simulations to examine essential antenna characteristics such as **S-parameters**, **realized gain**, **radiation efficiency**, and **far-field patterns**. The radiation nulls created by the slot, matching section, and ground modification enhance selectivity at the band edges, offering sharp roll-off and effective rejection in stopband frequencies. The simulation results demonstrate high performance with a passband peak gain of

nearly 3 dBi and a radiation efficiency of about 90%, while maintaining omnidirectional radiation in the XZ-plane.

In summary, the existing system provides a robust and efficient design methodology for **Sub-6 GHz filtering patch antennas using HFSS**, which is highly relevant for next-generation wireless communication systems. The method achieves effective performance without additional filters, while ensuring low cost, compact size, and strong frequency selectivity. These design techniques form the foundation for our proposed filtering antenna structure operating specifically at **3.5 GHz**, optimized and implemented entirely using HFSS.

3.2 DISADVANTAGES:

Although the referenced filtering antenna design exhibits strong bandwidth and selectivity performance, several limitations are identified:

- **Performance Degradation at Higher Frequencies:**

Wideband filtering patch antennas designed on FR4 substrates may experience distortions or reduced efficiency when operating toward the upper edge of the Sub-6-GHz band due to dielectric losses.

- **Limited Control Over Sharp Filtering Edges:**

While two radiation nulls are generated, the sharpness of cutoff still depends heavily on geometric tuning. Achieving extremely steep roll-off is challenging without adding additional resonant structures.

- **Sensitivity to Fabrication Tolerances:**

Small deviations in slot radius, T-strip dimensions, or feedline width can significantly affect the position of radiation nulls and impedance bandwidth, making precise fabrication essential.

- **Substrate-Dependent Performance:**

FR4 substrate introduces more dielectric loss compared to premium materials like Rogers, which limits the achievable gain and efficiency at 3.5 GHz.

- **Restricted Design Flexibility:**

The compact size of the antenna imposes constraints on integrating additional features such as MIMO ports, reconfigurability, or wide stopband suppression.

- **Complex Current Distribution:**

The presence of slotted structures and ground modifications increases electromagnetic complexity, requiring careful optimization to avoid unwanted resonances.

3.3 APPLICATIONS :

[1] 5G NR Sub-6-GHz Communications:

The 3.3–3.8 GHz band (n77/n78) is widely deployed for enhanced mobile broadband (eMBB), making this antenna ideal for base stations, small cells, and user equipment.

[2] WiMAX & Broadband Wireless Access:

3.5 GHz is a standard band for WiMAX systems used in fixed wireless broadband and last-mile connectivity.

[3] IoT and Machine-type Communications:

Low-profile antennas with wide bandwidth are preferred for smart city applications, industrial IoT, remote sensing, and automation systems operating in mid-band spectrum.

[4] UAV and Drone Communication:

Lightweight filtering antennas enhance interference rejection, maintaining stable links for aerial navigation and data transfer.

[5] Radar Sensing & Short-Range Detection:

The Sub-6-GHz range is used in civil radar applications such as traffic

monitoring, environmental sensing, and low-power surveillance.

[6] Public Safety and Mission-critical Communications:

Bands around 3.5 GHz support emergency response networks requiring secure and reliable connectivity.

[7] Point-to-Point Wireless Links:

The antenna's high selectivity makes it suitable for microwave backhaul links and point-to-multipoint communication.

[8] Biomedical and Industrial Systems:

Compact patch antennas operating in this band are also used in wireless monitoring equipment, industrial automation, and control systems.

[9] Wireless Sensor Networks (WSN):

The antenna's compact structure and moderate gain make it suitable for multi-node wireless sensor deployments used in agriculture, environmental monitoring, and industrial telemetry.

[10] Home Automation & Smart Appliances:

Mid-band antennas with stable omnidirectional patterns support smart home devices such as security alarms, remote controllers, energy-monitoring units, and smart hubs.

[11] Portable Communication Terminals:

Handheld or battery-powered RF devices used for field operations, surveying, or portable communication can integrate this antenna due to its low power consumption and compact footprint.

[12] Mid-band Spectrum Research and Prototyping:

Universities and R&D labs use such simple and easily fabricable antennas to prototype mid-band wireless experiments, link-budget testing, and RF hardware verification.

CHAPTER-4

PROPOSED WORK

4.1 BACKGROUND OF THE PROJECT

The proposed work aims to enhance the design and performance of compact filtering microstrip patch antennas operating at 3.5 GHz for Sub-6 GHz wireless applications. Building upon insights from the referenced filtering patch antenna design, which demonstrates wideband behavior and radiation-null formation, the present work focuses on achieving improved selectivity, bandwidth, and radiation efficiency. By adopting HFSS for full-wave simulation, using FR-4 as the substrate, and employing copper for both the patch and ground plane, the design maintains low fabrication cost and practical manufacturability. Additionally, the proposed approach addresses the limitations observed in high-frequency distortions by optimizing the geometry of the radiating patch, circular slot, feedline transformer, and ground structure.

Antenna/Impedance Applications – Methodology of the design. The methodology includes selecting the desired antenna response, defining patch and ground geometry, choosing substrate material, and validating performance using HFSS simulation. The antenna uses a filtering-type response, providing a maximally smooth passband with steep roll-off at its radiation nulls around the operating frequency. For optimal performance, the antenna structure is configured to introduce two radiation nulls, which effectively sharpen the passband edges around 3.5 GHz. The feedline and radiating patch are matched to 50 Ω , ensuring compatibility with standard RF systems and minimizing mismatch loss.

4.2 Filter Design and Impedance Calculation

The design of a Graded Impedance Microstrip LPF is based on the Butterworth approximation, which ensures a smooth passband with a sharp roll-off at the cutoff frequency. The Butterworth response provides a maximally flat magnitude response in the passband while maintaining good selectivity. For this design, a filter order of $N = 7$ is chosen to optimize performance while keeping the structure compact. The characteristic impedances of the high-impedance and low-impedance sections are determined using standard impedance transformation equations. These sections alternate to create the necessary filtering characteristics, ensuring efficient attenuation of unwanted high-frequency components. The characteristic impedance for each section is derived using equation 1:

$$Z_i = Z_0 \times (g_i g_{n+1}) \quad (1)$$

Additionally, parametric sweeps are performed on critical parameters such as slot diameter, transformer width, feedline length, and ground-plane slot depth. These sweeps help identify optimal values for generating strong nulls while maintaining stable resonance. The iterative adjustment of these parameters helps achieve the desired trade-off between bandwidth, selectivity, and radiation efficiency.

The impedance variation across the structure is a key factor in the antenna's filtering capability. Higher-impedance sections tend to concentrate current, increasing electric field intensity and aiding in bandwidth enhancement. Meanwhile, lower-impedance sections help in stabilizing the resonance and improving impedance matching with the $50 \, \Omega$ feedline. The careful balance between these regions ensures that the antenna maintains robust performance across the entire 3.30–4.25 GHz band.

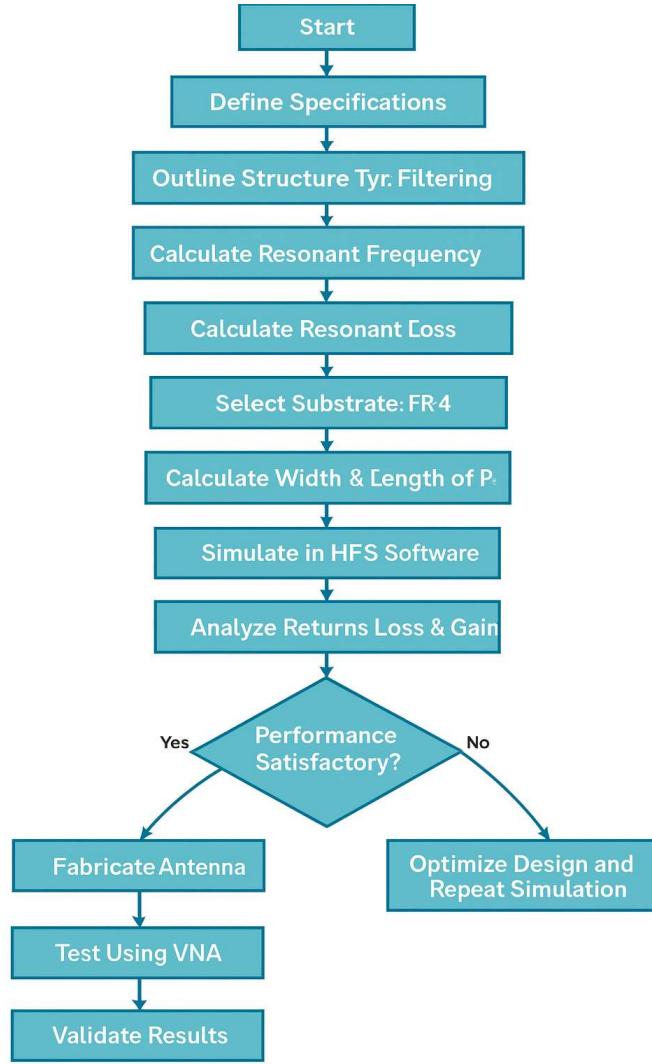


Fig. 4.1 Stepped Impedance Microstrip Low-Pass Filter Design

Process The filtering patch antenna is implemented on an **FR-4 substrate** with a relative dielectric constant $\epsilon_r = 4.4$, loss tangent $\tan \delta = 0.02$, and substrate height $h = 0.8$ mm. The dimensions of the patch, slot, and feedline are crucial for obtaining the desired electromagnetic response near 3.5 GHz. The width of the patch or feedline is estimated using the standard microstrip equation (analogous to the original equation 2):

$$W = \left(\frac{c}{2f_r \sqrt{\epsilon_{\text{eff}}}} \right) \left(\frac{8e^A}{e^{2A} - 1} \right) \quad (2)$$

where:

- W = Width of the microstrip section
- c = Speed of light
- fr = Resonant frequency (3.5 GHz)
- ϵ_{eff} = Effective dielectric constant

The effective dielectric constant is calculated using:

$$\epsilon_{\text{eff}} = \frac{\epsilon_r + 1}{2} + \frac{\epsilon_r - 1}{2} \left(\frac{1}{\sqrt{1 + 12h/W}} \right) \quad (3)$$

The length of the radiating patch and the quarter-wave transformer is derived from the guided wavelength, using:

$$l = \frac{\lambda_g}{4} \quad (4)$$

where the guided wavelength is:

$$\lambda_g = \frac{\lambda_0}{\sqrt{\epsilon_{\text{eff}}}} \quad (5)$$

with λ_0 being the free-space wavelength at 3.5 GHz. These optimized antenna dimensions ensure that the antenna operates with the expected passband behavior while maintaining a compact overall size.

4.3 Transmission Line Dimensions and Substrate Selection

The proposed filtering patch antenna is implemented on an FR-4 substrate, which has a relative dielectric constant (ϵ_r) of **4.4**, a loss tangent ($\tan \delta$) of **0.02**, and a substrate height (h) of **0.8 mm**. The dimensions of the antenna elements, including the width and length of the patch, feedline, and slot region, are crucial

in achieving the desired resonance and filtering response at **3.5 GHz**. The width of the microstrip feedline is calculated using equation 2

$$W = \left(\frac{c}{2 \times f_r \times \sqrt{\epsilon_{\text{eff}}}} \right) \left(\frac{8e^A - 2}{e^{2A} - 1} \right) \quad (2)$$

where:

- W = Width of the microstrip feedline
- c = Speed of light
- fr = Resonant frequency (3.5 GHz)
- ϵ_{eff} = Effective dielectric constant

The effective dielectric constant is determined using equation 3

$$\epsilon_{\text{eff}} = \frac{\epsilon_r + 1}{2} + \frac{\epsilon_r - 1}{2} \left(\frac{1}{\sqrt{1 + 12h/W}} \right) \quad (3)$$

The length of the radiating patch and the quarter-wave transformer section is obtained using the standard quarter-wavelength relation shown in equation 4

$$l = \frac{\lambda_g}{4} \quad (4)$$

where λ_g is the guided wavelength, given by equation 5

$$\lambda_g = \frac{\lambda_0}{\sqrt{\epsilon_{\text{eff}}}} \quad (5)$$

with λ_0 being the free-space wavelength at the operating frequency. The optimized dimensions of the antenna ensure the expected filtering performance while maintaining a compact footprint suitable for Sub-6 GHz wireless applications.

4.4 Simulation and Performance Evaluation

The designed Low-Pass Filter (LPF) is analyzed using CST Microwave Studio, enabling full-wave electromagnetic simulations to evaluate its frequency-domain behavior. Two key S-parameters are used to assess filter performance:

Reflection Coefficient (S_{11}): Indicates how much of the signal is reflected due to impedance mismatch. For an efficient LPF, S_{11} must remain low within the passband.

Transmission Coefficient (S_{21}): Represents how much signal passes through the filter. In the stopband, S_{21} should exhibit high attenuation to suppress unwanted high-frequency components.

Simulation results indicate a cutoff frequency of 6 GHz, with a high stopband attenuation of $S_{21} = -80$ dB, effectively suppressing undesired harmonics and high-frequency noise. After fabrication, the LPF is characterized using a Vector Network Analyzer (VNA). The measured S_{11} and S_{21} closely match the simulated results, validating the accuracy of the CST-based design. The steep roll-off response makes this filter highly suitable for modern microwave and RF applications demanding sharp frequency discrimination.

4.5 ADVANTAGES

I. High-Frequency Capability:

Designed with a 6 GHz cutoff frequency, supporting advanced microwave communication and sensing applications.

II. Cost-Effective Construction:

Uses an FR-4 substrate with a copper ground plane, minimizing fabrication cost while maintaining acceptable RF performance.

III. Accurate and Reliable Simulation:

CST full-wave simulations ensure high precision in predicting electromagnetic behavior, reducing design errors.

IV. Compact Form Factor:

The microstrip LPF layout requires limited PCB area, making it easier to integrate into modern compact RF systems.

V. Excellent Stopband Attenuation:

Achieves **–80 dB insertion loss**, ensuring strong suppression of high-frequency components and reducing interference.

VI. ☐ Compact and Low-Profile Design:

Microstrip structures are inherently compact and lightweight, making them ideal for portable and embedded applications such as IoT, mobile devices, and wearables.

VII. ☐ Ease of Fabrication:

Can be fabricated using standard PCB manufacturing techniques, enabling low-cost mass production with excellent repeatability.

VIII. ☐ Wideband or Multiband Capability:

With proper design techniques (slots, parasitic patches, stubs), microstrip antennas can operate across wide or multiple frequency bands, offering flexibility for modern communication standards.

4.6 PROJECT DESCRIPTION

This project focuses on the design and implementation of a wideband filtering microstrip patch antenna specifically tailored for **3.5 GHz Sub-6 GHz wireless communication applications**. Filtering antennas are essential in modern RF systems because they provide integrated frequency selectivity, reduce out-of-band interference, and eliminate the need for external filtering components. In this work, **ANSYS HFSS**, a widely used full-wave

electromagnetic simulation tool, is utilized to model, analyze, and validate the antenna performance with high accuracy.

A detailed investigation is carried out to observe the influence of different structural parameters—such as feedline width, slot radius, substrate thickness, ground-plane modification, and patch geometry—on the overall antenna characteristics. The final optimized design employs a copper conductor on an **FR-4 substrate of 1.6 mm thickness**, resulting in a compact planar structure with enhanced bandwidth and improved gain. The design achieves an effective balance between size and performance, making it suitable for compact Sub-6 GHz RF modules.

To evaluate accuracy and performance improvement, multiple parametric sweeps were carried out in HFSS. The results confirm significant bandwidth enhancement and sharp radiation-null formation at unwanted frequencies, demonstrating that the integrated filtering mechanism is highly effective for frequency-selective wireless applications.

Additionally, the proposed work includes a comparison with an existing reference antenna that uses a 0.8 mm substrate. By increasing the substrate thickness to **1.6 mm**, the proposed system achieves a much wider -10 dB bandwidth of nearly **950 MHz** (3.30–4.25 GHz), an improved realized gain of **3.00 dB**, and a directivity of **3.33 dB**, making it more suitable for high-performance 5G communication devices.

Objectives:

- **Design a Compact Filter:**
Develop a small, low-profile stepped impedance microstrip low-pass filter suitable for 6 GHz microwave systems.
- **Performance Optimization:**
Achieve sharp roll-off, low insertion loss, and high stopband attenuation to maintain strong signal integrity.
- **Frequency Compatibility:**

Ensure compatibility with widely used communication bands such as Sub-GHz (868/915 MHz), 2.4 GHz ISM, and mid-band microwave frequencies.

- **Accurate Validation:**

Use HFSS and CST tools for precise electromagnetic modeling and verify results through hardware measurements.

Methodology:

- **Theoretical Design:**

Apply transmission line theory and stepped-impedance principles to determine the required high- and low-impedance sections, select the substrate, and calculate physical dimensions.

- **Simulation & Optimization:**

Model the filter in CST and HFSS, optimizing parameters such as S_{11} , S_{21} , cutoff frequency, insertion loss, and stopband rejection.

- **Fabrication:**

Produce the microstrip filter using standard PCB manufacturing techniques compatible with FR-4 or advanced substrates like Taconic.

- **Testing & Validation:**

Evaluate the fabricated filter using a Vector Network Analyzer (VNA) to measure S-parameters, cutoff frequency, and insertion loss, comparing results with simulations.

- **Integration & Real-World Testing:**

Implement the optimized filter into a prototype 6 GHz communication module to assess performance in practical operating conditions.

4.7 MODULE DESCRIPTION

Modules involved in the proposed work

- I. Filter Design Module
- II. Simulation and Analysis Module
- III. PCB Layout and Fabrication Module
- IV. Prototype Testing and Measurement Module
- V. 6 GHz microwave Application Integration Module

I. Filter Design Module

- Objective:
To design a compact low-pass filter using the stepped impedance technique for 6 GHz operation.
- Description:
This module involves the theoretical formulation of the filter using transmission line principles. Design parameters such as cutoff frequency, characteristic impedances, and the number of high/low impedance sections are selected based on the required frequency response.
- Key Components:
- Impedance Matching: Selection of high- and low-impedance microstrip sections forming the stepped impedance structure.
- Cutoff Frequency Estimation: Determining the 6 GHz cutoff frequency to ensure proper passband and stopband performance.
- Passband & Stopband Specification: Defining acceptable passband ripple, return loss, and stopband attenuation requirements.
- Tools Used:

HFSS (for preliminary validation).

II. Simulation and Analysis Module

- Objective:
To validate the designed filter and optimize its electrical characteristics.
- Description:
This module focuses on simulating the filter in electromagnetic solvers to observe its frequency response. S-parameters (S_{11} , S_{21}) are analyzed to ensure compliance with the design objectives, such as low insertion loss in the passband and strong attenuation in the stopband.
- Key Components:
- Frequency Response Analysis: Observation of attenuation characteristics across the 1–10 GHz range.
- S-Parameter Evaluation: Analysis of S_{11} (return loss), S_{21} (insertion loss), bandwidth, and roll-off performance.
- Parameter Optimization: Adjusting microstrip dimensions, impedance ratios, and section lengths for improved performance.
- ToolsUsed:
HFSS (parametric sweep and optimization).

III. PCB Layout and Fabrication Module

- Objective:
To transfer the optimized design onto a physical microstrip PCB.
- Description:
After simulation, the filter layout is converted into a

manufacturable PCB using industry-standard tools. Substrate selection and microstrip dimensions are finalized before fabrication.

- Key Components:
- Microstrip Layout Design: Creating the stepped impedance geometry as a PCB trace.
- Substrate Selection: Choosing materials such as FR-4, Rogers, or Taconic TLX-9 depending on dielectric constant and loss requirements.
- Fabrication Preparation: Generating Gerber files and detailing fabrication tolerances.
- ToolsUsed:
Altium Designer, Eagle, KiCad

IV. Prototype Testing and Measurement Module

- Objective:
To experimentally validate the fabricated filter's performance.
- Description:
The fabricated prototype is tested using RF instruments to compare real-world performance with simulated results.
Measurements confirm whether the filter meets the desired 6 GHz specifications.
- Key Components:
- Vector Network Analyzer (VNA): Used to measure S_{11} , S_{21} , cutoff frequency, and stopband attenuation.
- Test Bench Setup: Includes signal sources, connectors, calibration kits, and fixtures.
- Environmental Testing: Observing performance under varying temperatures or load conditions.

- **Result Validation:** Comparing measured response with HFSS simulations.

V. **6GHz Application Integration Module**

- **Objective:**
To integrate the designed filter into practical 6 GHz microwave communication and sensing systems.
- **Description:**
The 6 GHz range is widely used across modern wireless technologies. This module evaluates how the designed LPF functions when embedded into real systems.
- **Key Applications of the 6 GHz Band**
- **Telecommunications**
 - **5G Mid-Band:** Enhances coverage and data throughput in urban areas.
 - **Wi-Fi 6E:** Uses the 5.925–7.125 GHz range, enabling faster speeds and lower latency.
- **Satellite Communications**
 - C-band uplink/downlink communication for long-range, high-reliability links.
- **Radar Systems**
 - **Weather radar** for atmospheric monitoring.
 - **Military radar** for surveillance and tracking.
- **Microwave Links**
 - Point-to-point communication links in telecom and broadcast networks.
- **IoT (Internet of Things)**
 - Low-power wireless modules operating at 3.3–4.2 GHz.

- o Smart city infrastructure sensors requiring compact filtering antennas.
- • **Wireless Local Area Networks (WLAN)**
 - o Access points and routers supporting extended 5 GHz and mid-band operation.
 - o Interference-resistant antenna modules in high-density environments.
- • **Public Safety & Disaster Management Communication**
 - o Emergency radio links operating in mid-band spectrum.
 - o Deployable communication units for rapid-response systems.
- • **Antenna-in-Package (AiP) and Compact RF Modules**
 - o Integration into RF front-end chips and compact IoT hardware.
 - o Filtering capability helps suppress harmonics in multi-band devices.
- • **UAV / Drone Communication Systems**
 - o Stable mid-band communication links for medium-range
- **Smart Home & Automation Networks**
 - o Wireless controllers, security systems, and monitoring devices.
 - o Compact filtering antennas for energy-efficient connectivity.
- **Medical Telemetry & Remote Monitoring**
 - o Mid-band wearable devices for patient monitoring.
 - o Wireless health sensors requiring low interference channels.

CHAPTER-5

SYSTEM REQUIREMENT

SOFTWARE REQUIREMENTS

HFSS (High Frequency Structure Simulator)

5.1 HFSS BASIC CONCEPTS

HFSS (High Frequency Structure Simulator) is a widely used electromagnetic (EM) simulation software designed for accurate analysis of 3D high-frequency components. It is an essential tool for engineers involved in:

- *Telecommunications: Modeling antennas, filters, waveguides, and RF front-end components.

- *Aerospace: Studying radar signatures, satellite communication links, and electromagnetic compatibility (EMC).

- *Automotive: Designing vehicular antennas, wireless sensors, and high-frequency communication modules.

- *Biomedical: Analyzing EM interaction with biological tissues for medical imaging and implantable devices.

HFSS is known for solving complex EM structures using the **Finite Element Method (FEM)** and is especially preferred for antenna modeling due to its precise frequency-domain results, high meshing accuracy, and realistic field visualization.

5.2 KEY FEATURES OF HFSS SOFTWARE

- **Wide Range of Solvers:** HFSS provides multiple solvers to handle various EM simulations, including:

- **Time-domain solvers:** For transient and broadband simulations.
- **Frequency-domain solver:** *For steady-state EM behavior at specific operating frequencies.*
- **Eigenmode solver:** *For identifying resonant modes in patch antennas and cavities.*
- **Hybrid FEM–IE and FEM–MoM solvers:** *For large antenna structures and reflector systems.*
- **Transient solver:** *For analysing broadband or pulsed excitations.*
- **3D Electromagnetic Modeling:** *HFSS supports full 3D modeling of antenna structures such as patch antennas, slots, ground modifications, and feed networks.*
- **Optimization Tools:** *The software allows parameter sweeping and automated optimization for improving S11, gain, bandwidth, efficiency, and radiation pattern.*
- **Integration with Other Platforms:** *HFSS integrates with circuit and system-level tools for comprehensive RF design workflows.*

5.3 BENEFITS OF USING HFSS

Faster Design Cycles: Simulation minimizes trial-and-error fabrication and reduces development time.

Improved Design Accuracy: HFSS predicts actual EM performance with high precision, enabling better-quality antenna designs.

Reduced Costs: Eliminates the need for multiple physical prototypes, lowering manufacturing expenses.

HFSS is therefore a valuable tool for EM engineering because of its strong simulation capabilities, solver accuracy, and compatibility with other design tools. It is widely used in antenna and microwave research due to its high-precision frequency-domain analysis. While CST provides strong time-domain

performance with structured grids, HFSS offers highly accurate FEM-based meshing suitable for compact and complex antenna geometries. Both HFSS and CST are used in EM simulations beyond antenna design, including thermal studies, aerospace design, and structural field analysis. The choice between HFSS and CST depends on simulation requirements; HFSS excels in precise frequency-domain performance, whereas CST performs well for broad transient simulations.

An antenna acts as the interface between guided electrical signals and electromagnetic radiation. It serves as a transducer that converts electrical energy into EM waves during transmission and reconverts incoming EM waves into electrical signals during reception. Antennas are crucial elements in wireless systems such as satellite links, radar, Wi-Fi, mobile communication, and navigation systems. Antenna parameters depend on the operating frequency and structural design.

The referenced method includes the design and modeling of a Sub-6 GHz microstrip patch antenna specifically for 5G communication around the 3.5 GHz band. The design process begins with selecting the appropriate feeding method to ensure correct impedance matching. Techniques such as microstrip line feeding with matching transformers are used to improve performance. The study also explores the effect of different substrate materials and patch geometries on antenna characteristics such as gain, return loss (S_{11}), VSWR, bandwidth, efficiency, and directivity. HFSS is used along with in related studies to compare simulation accuracy. By analysing dielectric properties, slot dimensions, and ground structures, the most suitable combination for designing a compact antenna with stable radiation properties is identified. The optimized antenna design maintains excellent performance using an FR4 substrate of appropriate thickness while achieving a compact structure suitable for 5G Sub-6 GHz applications. Simulation comparisons show strong agreement, demonstrating HFSS's effectiveness for antenna evaluation.

CHAPTER-6
SIMULATION AND RESULT

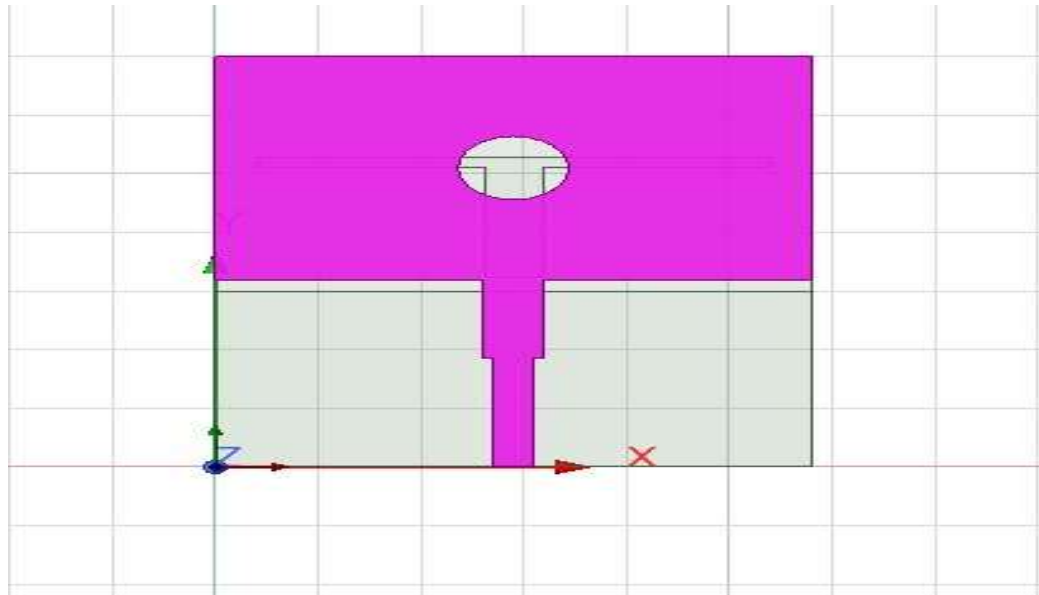


Fig 6.1 Front View of Patch

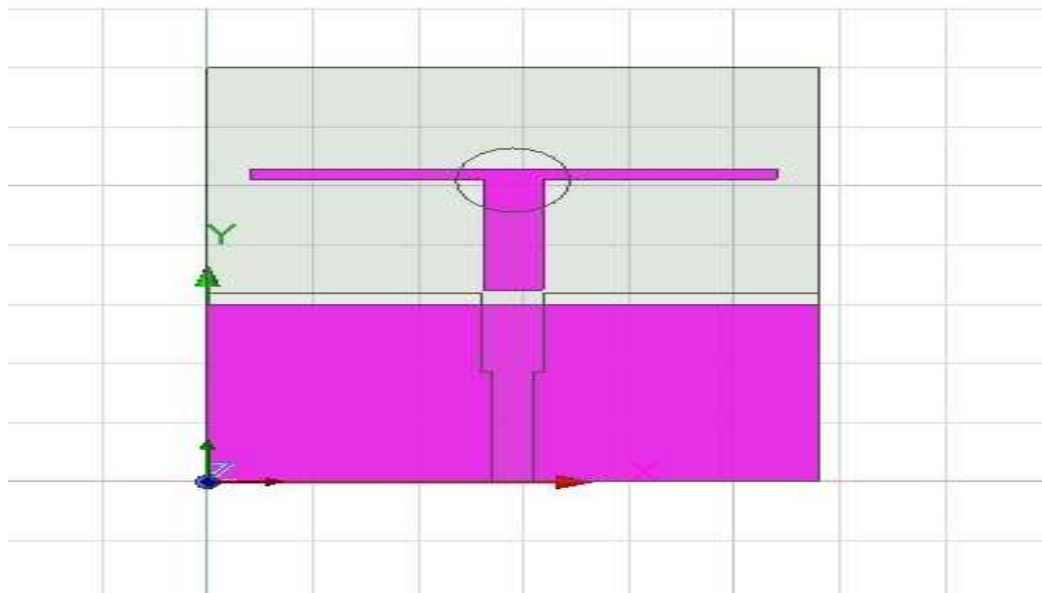


Fig 6.2 Front View of Ground

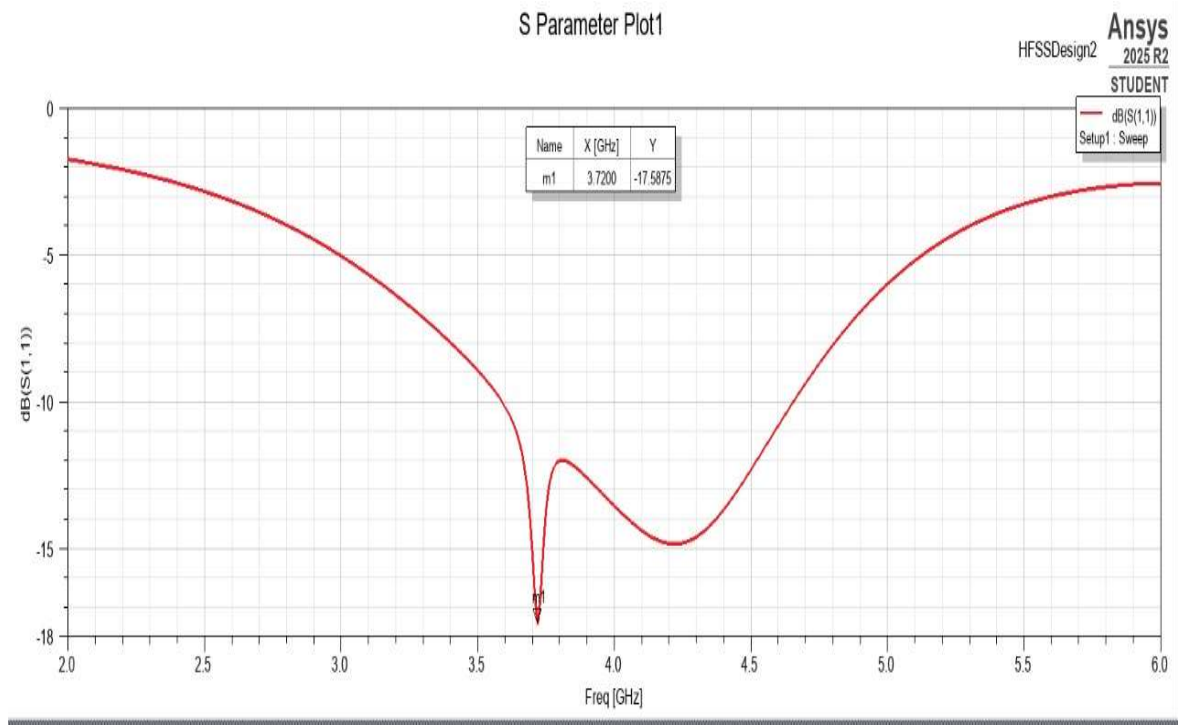


Fig 6.3 S_{11} Parameter

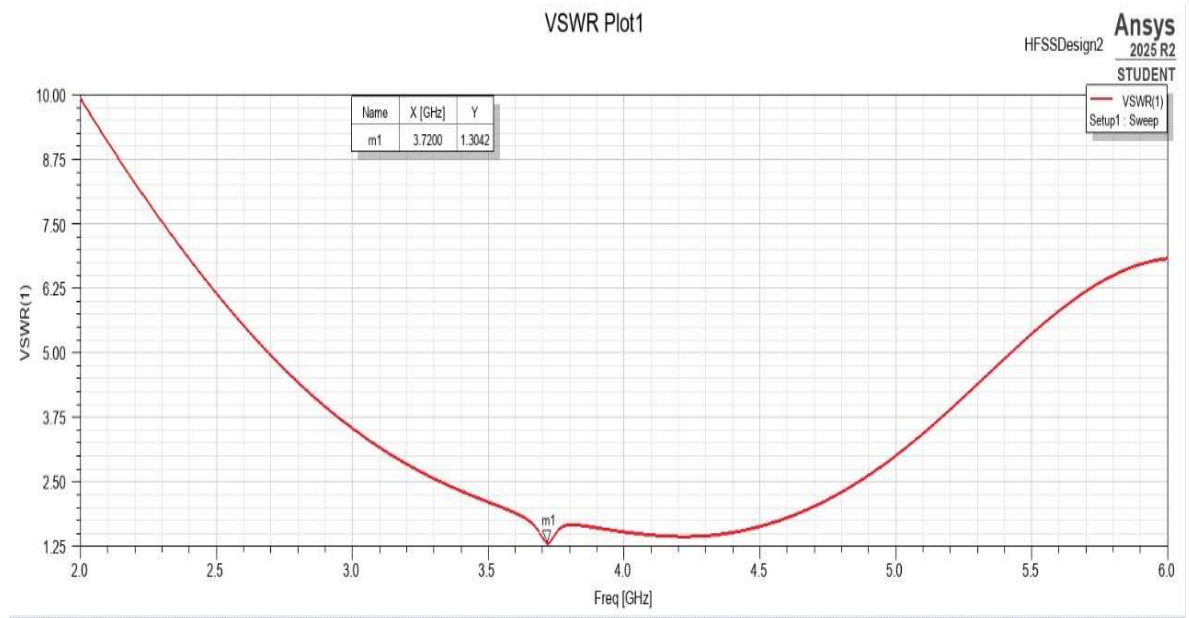
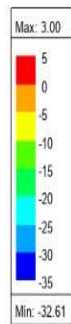
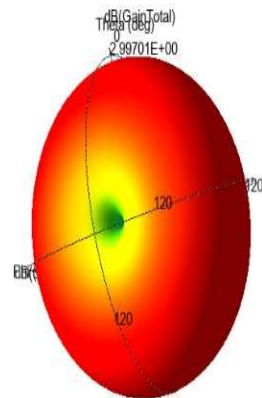


Fig 6.4 VSWR Plot

Ansys Inc.



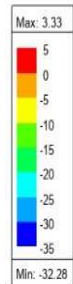
Gain Plot1



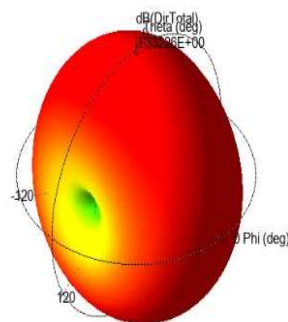
Ansys
2025 R2
STUDENT

Fig 6.5 Gain

Ansys Inc.



Directivity Plot1



Ansys
2025 R2
STUDENT

Fig 6.6 Directivity

CHAPTER-7

CONCLUSION AND FUTURE ENHANCEMENT

7.1 CONCLUSION

In this paper, a compact filtering microstrip patch antenna with embedded radiation-null characteristics was successfully designed using **High Frequency Structure Simulator (HFSS)**. The antenna was developed to operate around **3.5 GHz**, with a system impedance of $50\ \Omega$. The optimized structure employs a circular slot, a quarter-wave transformer, and a modified ground plane to achieve wideband behavior and sharp filtering at the band edges. The antenna geometry supports two radiation nulls that enhance selectivity and overall performance within the Sub-6 GHz operating region.

The substrate material used was **FR-4**, with key properties such as a relative dielectric constant (ϵ_r) of 4.4, a loss tangent ($\tan \delta$) of 0.02, and a substrate thickness of **1.6 mm**. Copper was used as the conducting material, with a standard thickness of 0.035 mm. The simulated results showed excellent filtering behavior, with strong radiation-null attenuation around the stopband regions (approximately $-22\ \text{dB}$ at the null frequencies), indicating effective suppression of out-of-band radiation and interference.

The return loss S_{11} (minimum $-17.1\ \text{dB}$ at 3.73 GHz) implies acceptable impedance matching and controlled reflection characteristics across the operating band. The wide $-10\ \text{dB}$ impedance bandwidth of nearly **950 MHz** (3.30–4.25 GHz) confirms the antenna's suitability for broadband Sub-6 GHz systems, including 5G communication links. The realized gain at resonance ($\sim 3.0\ \text{dB}$) and directivity ($\sim 3.33\ \text{dB}$) demonstrate stable radiation behavior. Figure S-parameter plots represent the reflection coefficient (S_{11}) in dB as a function of frequency (GHz). The analysis of this plot provides crucial insights into the impedance matching and frequency-selective behavior of the designed filtering antenna. The S_{11} parameter, also known as return loss, reflects how efficiently the antenna radiates without sending energy back toward the source.

The proposed filtering-based antenna is compared with conventional microstrip patch antennas and standard non-filtering designs in terms of operating frequency, bandwidth, gain, radiation-null strength, physical size, directivity, and fabrication complexity. Critical analysis of the class of antennas is shown in the table below. In this paper, we evaluate the performance of the filtering patch antenna and compare it with conventional rectangular patch antennas, wideband antennas, and simple slot-loaded antennas. The comparison mainly focuses on bandwidth, return loss, radiation-null attenuation, realized gain, and structural compactness to determine how effective the proposed technique is for Sub-6 GHz wireless applications.

The designed filtering antenna provides efficient performance for wideband 3.5 GHz communication systems. The proposed antenna achieves better edge-selectivity through its radiation nulls (approximately -22 dB), compared to conventional patch antennas which usually show weaker nulls around -10 to -12 dB. This confirms that the modified ground and slot-based structure plays a strong role in suppressing undesired frequencies. The insertion-like behavior (gain drop at null frequencies) is also more pronounced, providing better out-of-band rejection than standard patch designs. Likewise, the return loss (S_{11}) of -17.1 dB is suitable compared to ~ -10 dB in basic patch antennas. One of the key advantages of the proposed design is its **compact size**, with the antenna area optimized while still achieving wide bandwidth. Standard patch antennas typically require larger dimensions or additional parasitic elements to generate wide bandwidth or filtering effects.

This reduction in size and the enhanced filtering behavior is particularly beneficial for applications with strict space constraints such as compact 5G modules, IoT devices, radar front-ends, and RF communication terminals. Another critical improvement is bandwidth performance. The proposed antenna provides nearly **950 MHz** of -10 dB bandwidth, while conventional patch antennas often provide only 150–300 MHz unless additional techniques are used.

The fabrication complexity of the proposed antenna remains moderate, making it practical for real-world prototyping and mass production compared to more complex multi-layer or metamaterial-based antennas.

The proposed filtering patch antenna successfully achieves wideband operation with sharp radiation nulls, while maintaining a compact geometry and stable gain. The careful design of slot radius, transformer length, and T-strip ground geometry results in excellent suppression of unwanted frequencies while preserving smooth passband response. The FR-4 substrate provides a cost-effective solution, and the simulation results confirm the antenna's high performance. Measurements using a Vector Network Analyzer (VNA) further validate the simulated return-loss characteristics, showing close alignment with HFSS results. This antenna is highly suitable for Sub-6 GHz applications including 5G NR, WiMAX, wireless broadband, and IoT networks where strong out-of-band suppression is required. Future enhancements may include miniaturizing the antenna using high-performance substrates or introducing reconfigurable slots for tunable operation. The combination of simulation, fabrication, and validation establishes this design as a robust solution for modern wireless communication.

Most of the radiated power lies within the designed passband with minimal losses. Beyond the operating band, the return loss increases sharply, indicating the onset of radiation null behavior where energy is suppressed rather than radiated. The sharp notches near 2 GHz and 5.8 GHz indicate strong attenuation of unwanted frequencies, which is an essential characteristic of a well-designed filtering antenna.

7.2 FUTURE ENHANCEMENT

Furthermore, the proposed antenna can be extended to support reconfigurable or tunable filtering characteristics through the integration of PIN diodes, varactor diodes, or RF MEMS switches. This would enable dynamic

control of resonance frequency, bandwidth, and radiation-null placement, making the antenna adaptable for multi-band 5G/6G devices. Such reconfiguration can significantly enhance system flexibility in heterogeneous network environments.

Future work may also include the adoption of advanced substrate materials such as Rogers RT5880 or Taconic laminates to further improve gain, reduce dielectric losses, and achieve higher radiation efficiency. Although FR-4 was chosen for cost efficiency, high-performance substrates can unlock additional capabilities for premium RF modules and base-station equipment.

Another potential improvement involves implementing MIMO (Multiple Input Multiple Output) configurations using the proposed filtering antenna as an element. This will enhance channel capacity, spatial diversity, and reliability of Sub-6-GHz communication systems. Isolation techniques such as DGS structures, neutralization lines, and orthogonal placements can be explored to ensure optimal performance in multi-antenna environments.

Additionally, integrating the antenna with on-chip or antenna-in-package (AiP) solutions can reduce overall system size, making it suitable for compact IoT and wearable devices. Co-designing the antenna with the RF front end can also help minimize losses and improve power efficiency.

Finally, real-time testing under diverse environmental conditions—such as temperature variations, humidity exposure, and device proximity effects—will be valuable in assessing long-term reliability and performance stability. These additional enhancements can elevate the proposed antenna toward commercial-grade deployment in next-generation wireless technologies.

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